

Homework 3: Filter Banks

1 Filter Banks and Product Filters

Consider a two-channel filter bank with analysis filters (H_0, H_1) and synthesis filters (G_0, G_1) such that

$$\begin{aligned} H_0(z) &= 1 \\ G_0(z) &= \frac{1}{2} + z^{-1} + \frac{1}{2}z^{-2} \\ H_1(z) &= G_0(-z) \\ G_1(z) &= -H_0(-z). \end{aligned}$$

Prove that this system is PR. What is the total delay of the system? What is the product filter $P_0(z)$? Is the product filter half-band?

Solution. Clearly the no-aliasing condition is satisfied. For the no-distortion condition, we have that

$$G_0(z)H_0(z) + G_1(z)H_1(z) = \frac{1}{2}(1 + z^{-1})^2 + (-1)\frac{1}{2}(1 - z^{-1})^2 = 2z^{-1}$$

and so the total delay of the system is $L = 1$. The product filter is

$$P_0(z) = G_0(z)H_0(z) = \frac{1}{2}(1 + z^{-1})^2$$

which is clearly a half-band filter centered at 1.

2 Polyphase Representations of Filter Banks

- (a) Find the analysis polyphase matrix $\mathbf{H}_p(z)$ for the analysis filters $H_0(z) = 1$ and $H_1(z) = \frac{1}{16}(-1 + 9z^{-2} - 16z^{-3} + 9z^{-4} - z^{-6})$.

Solution.

$$\mathbf{H}_p(z) = \begin{bmatrix} 1 & 0 \\ \frac{1}{16}(-1 + 9z^{-1} + 9z^{-2} - z^{-3}) & -z^{-1} \end{bmatrix}$$

- (b) Find the synthesis polyphase matrix $\mathbf{G}_p(z)$ for the synthesis filters $G_0(z) = \frac{1}{16}(-1 + 9z^{-2} + 16z^{-3} + 9z^{-4} - z^{-6})$ and $G_1(z) = -1$.

Solution.

$$\mathbf{G}_p(z) = \begin{bmatrix} z^{-1} & 0 \\ \frac{1}{16}(-1 + 9z^{-1} + 9z^{-2} - z^{-3}) & -1 \end{bmatrix}$$

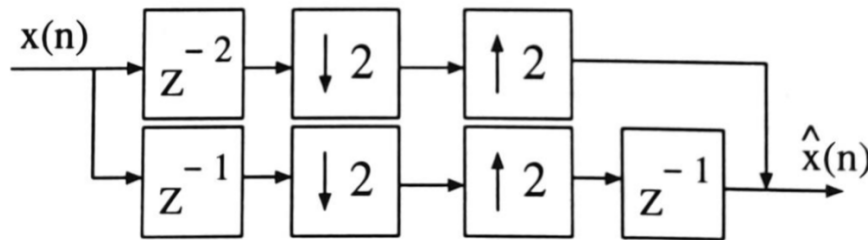
- (c) Verify that $\mathbf{G}_p(z)\mathbf{H}_p(z) = z^{-K}\mathbf{I}$ for the matrices you found in (a) and (b) and determine K .

Solution. Multiplying the two matrices

$$\mathbf{G}_p(z)\mathbf{H}_p(z) = \begin{bmatrix} z^{-1} & 0 \\ 0 & z^{-1} \end{bmatrix} = z^{-1}\mathbf{I}.$$

Therefore, $K = 1$, which implies that the total delay of the system is $L = 2K + 1 = 3$.

- (d) Find the polyphase matrices $\mathbf{H}_p(z)$ and $\mathbf{G}_p(z)$ for the following system:



Is this system PR?

Solution. The analysis bank has $H_0(z) = z^{-2}$ and $H_1(z) = z^{-1}$. This leads to the polyphase matrix

$$\mathbf{H}_p(z) = \begin{bmatrix} z^{-1} & 0 \\ 0 & 1 \end{bmatrix}.$$

Meanwhile, the synthesis bank has $G_0(z) = 1$ and $G_1(z) = z^{-1}$. This leads to the polyphase matrix

$$\mathbf{G}_p(z) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Clearly the system is PR with total delay 2, i.e., $\hat{x}[n] = x[n - 2]$. To see this, observe that we can factor out one delay z^{-1} to the very beginning of the system and we are then left with the cascade of a single delay with the system from the beginning of the lecture 8 notes (which we proved corresponded to one delay). Therefore the whole system corresponds to a delay of $L = 2$.

- (e) Given a PR filter bank with filters (H_0, H_1, G_0, G_1) , suppose that we replace the analysis filters with $\hat{H}_0(z) = H_0(z)$ and $\hat{H}_1(z) = z^{-2}H_1(z)$. How are the polyphase matrices $\mathbf{H}_p(z)$ and $\hat{\mathbf{H}}_p(z)$ related? Is the new system PR?

Solution. We have that

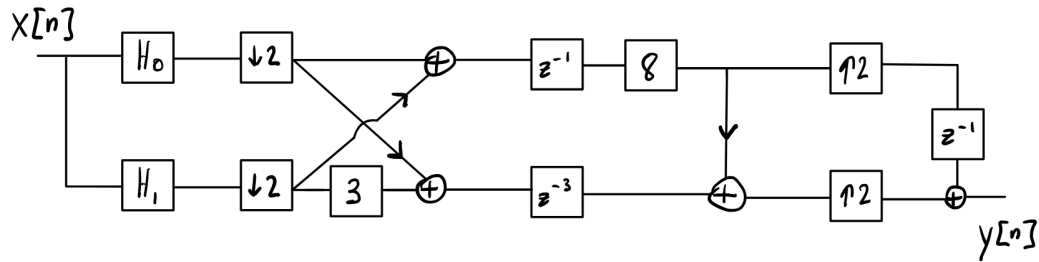
$$\hat{\mathbf{H}}_p(z) = \begin{bmatrix} 1 & 0 \\ 0 & z^{-1} \end{bmatrix} \mathbf{H}_p(z).$$

To guarantee PR of the new system, we would have to delay the synthesis polyphase matrix to account for the delay in the high-pass channel. So, if we modify the synthesis back via, e.g.,

$$\hat{\mathbf{G}}_p(z) = \mathbf{G}_p(z) \begin{bmatrix} z^{-J} & 0 \\ 0 & z^{-J+1} \end{bmatrix},$$

the new system would be PR.

- (f) Consider the two-channel filter bank:



Find the synthesis polyphase matrix $\mathbf{G}_p(z)$ and the synthesis filters (G_0, G_1) .

Solution.

$$\mathbf{G}_p(z) = \begin{bmatrix} 8z^{-1} & 8z^{-1} \\ 8z^{-1} + z^{-3} & 8z^{-1} + 3z^{-3} \end{bmatrix}.$$

Therefore,

$$\begin{aligned} G_0(z) &= 8z^{-2} + z^{-6} + 8z^{-3} \\ G_1(z) &= 8z^{-2} + 3z^{-6} + 8z^{-3}. \end{aligned}$$

3 Orthogonal Filter Banks

Multiply out $\mathbf{H}(z) = \mathbf{R}_1 \mathbf{\Lambda}(z) \mathbf{R}_0$, where $\mathbf{R}_k$, $k = 1, 2$, is a rotation matrix by angle $-\theta_k$ and $\mathbf{\Lambda}(z)$ is defined as in lecture. Suppose that this is a polyphase matrix. What are the filter coefficients $d[n]$ that come from the even and odd polyphases from the second row of $\mathbf{H}(z)$? Let $D(z)$ denote the z -transform of $d[n]$. If $D(1) = 0$ prove that $\theta_1 + \theta_0 = -\frac{\pi}{4} + n\pi$, $n \in \mathbb{Z}$.

Solution. Multiplying out the matrices gives

$$\begin{aligned} \mathbf{H}(z) &= \mathbf{R}_1 \mathbf{\Lambda}(z) \mathbf{R}_0 = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 \\ \sin \theta_1 & \cos \theta_1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & z^{-1} \end{bmatrix} \begin{bmatrix} \cos \theta_0 & -\sin \theta_0 \\ \sin \theta_0 & \cos \theta_0 \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta_0 \cos \theta_1 - z^{-1} \sin \theta_0 \sin \theta_1 & -\sin \theta_0 \cos \theta_1 - z^{-1} \cos \theta_0 \sin \theta_1 \\ \cos \theta_0 \sin \theta_1 + z^{-1} \sin \theta_0 \cos \theta_1 & -\sin \theta_0 \sin \theta_1 + z^{-1} \cos \theta_0 \cos \theta_1 \end{bmatrix}. \end{aligned}$$

This gives the filter coefficients for $d[n]$ as

$$\{\cos \theta_0 \sin \theta_1, \quad -\sin \theta_0 \sin \theta_1, \quad \sin \theta_0 \cos \theta_1, \quad \cos \theta_0 \cos \theta_1\}$$

From the condition $D(1) = 0$ (which means that $D(e^{j\omega})|_{\omega=0} = 0$), we obtain

$$\begin{aligned} \cos \theta_0 \sin \theta_1 - \sin \theta_0 \sin \theta_1 + \sin \theta_0 \cos \theta_1 + \cos \theta_0 \cos \theta_1 &= 0 \\ \sin(\theta_0 + \theta_1) + \cos(\theta_0 + \theta_1) &= 0 \\ \tan(\theta_0 + \theta_1) &= -1 \\ \theta_0 + \theta_1 &= -\frac{\pi}{4} + n\pi, \quad n \in \mathbb{Z}. \end{aligned}$$